ELTON MAYO

HAWTHORNE EXPERIMENT

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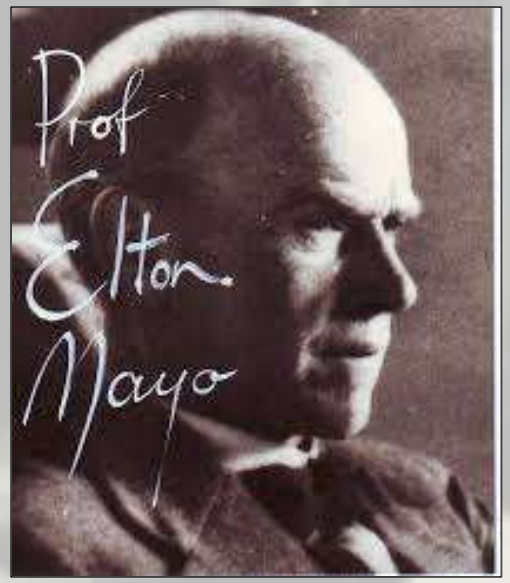


- George Elton Mayo (1880–1949) was an Australian born psychologist, researcher and organizational theorist
- Mayo is known as the founder of the Human Relations Movement
- The research he conducted under the rubric of the Hawthorne Studies in the late 1920s and early 1930s showed the importance of groups in affecting the behavior of individuals at work.
- He carried out a number of investigations to look at ways of improving productivity

WESTERN ELECTRIC COMPANY

- Gray and Barton, a telephone industry supply company founded in 1869 by Elisha Gray and Enos Barton

- In 1872, the company changed its name to the Western Electric Manufacturing Company

- In 1881, when the annual sales had already grown to nearly $1 million, the firm was purchased by the American Bell Telephone Company, it was renamed the Western Electric Company and became Bells manufacturing arm

HAWTHORNE EFFECT

- The Hawthorne Effect is the idea that “behavior during the course of an experiment can be altered by a subject’s awareness of participating in an experiment”

- The initial Hawthorne effect took place in the Hawthorne plant of Western Electric Company in the 1920’s and 1930’s

- The studies were composed of many long “investigations into the importance for work behavior and attitudes of a variety of physical, economic, and social variables.”

HAWTHORNE EXPERIMENT

- The Hawthorne experiment were first conducted in November, 1924 at Western Electric Company's Hawthorne plant in Chicago
- The initial tests were sponsored by The National Research Council (NRC) of the National Academy of Sciences
- In 1927, a research team from Harvard Business School was invited to join the studies after the illumination test drew unanticipated results
- A team of researchers led by George Elton Mayo from the Harvard Business School carried out the studies
- (General Electric originally contributed funding, but they withdrew after the first trial was completed)

4 PARTS OF HAWTHORNE EXPERIMENT

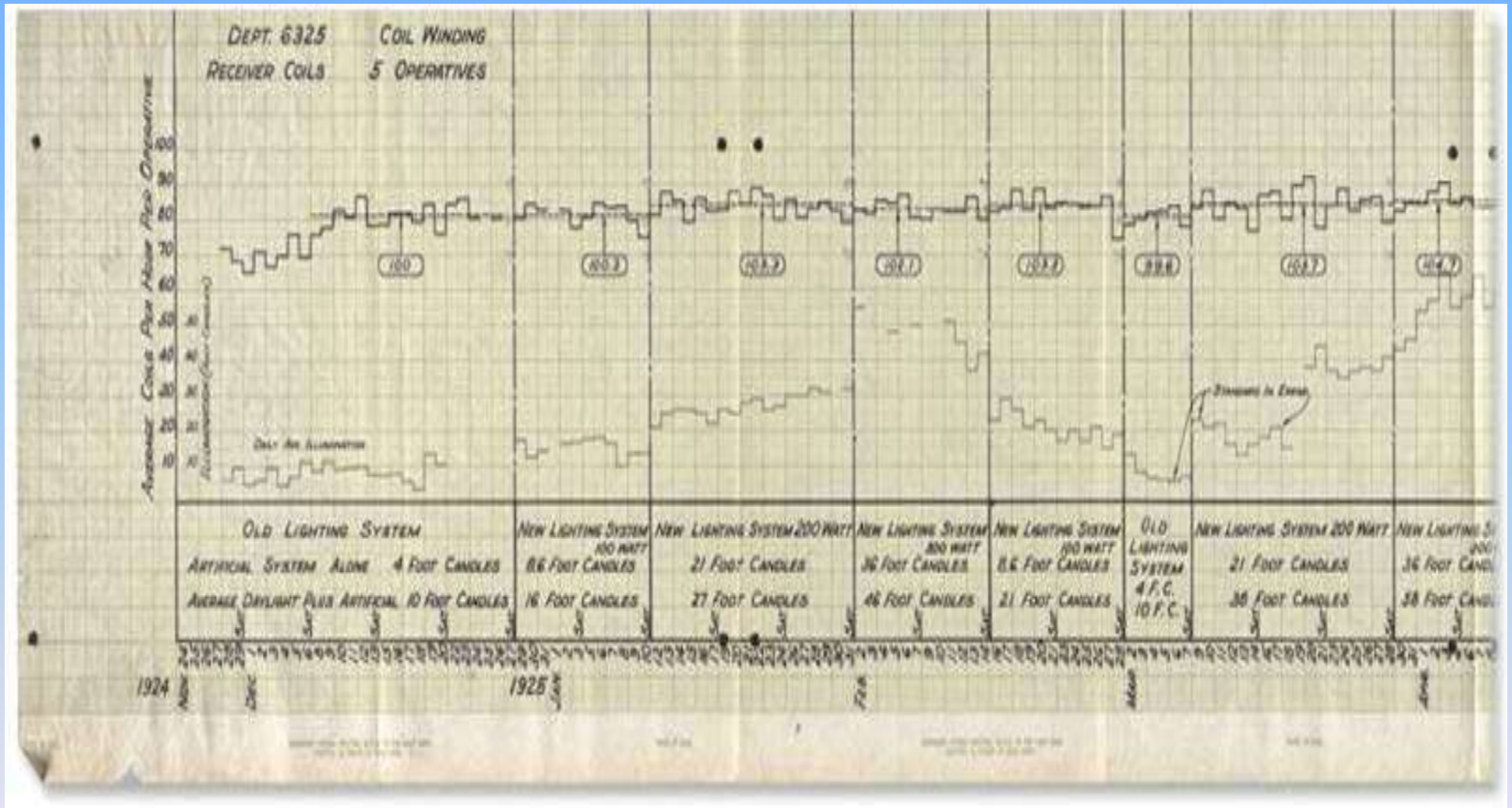
I. Part I - Illumination Experiments (1924-27)
II. Part II - Relay Assembly Test Room Study (1927-1929)
III. Part III - Mass Interviewing Programme (1928-1930)
IV. Part IV - Bank Wiring Observation Room Experiment (1932)

PART I - ILLUMINATION EXPERIMENTS (1924-27)

- These experiments were performed to find out the effect of different levels of illumination (lighting) on productivity of labour. The brightness of the light was increased and decreased to find out the effect on the productivity of the test group. Surprisingly, the productivity increased even when the level of illumination was decreased. It was concluded that factors other than light were also important.

- Result :
  - Higher worker productivity and satisfaction at all light levels
  - Worker productivity was stopped with the light levels reached moonlight intensity.
- Conclusions:
  - Light intensity has no conclusive effect on output
  - Productivity has a psychological component

ILLUMINATION STUDY



PART II – RELAY ASSEMBLY TEST ROOM STUDY (1927–1929)

- Under these test two small groups of six female telephone relay assemblers were selected. Each group was kept in separate rooms. From time to time, changes were made in working hours, rest periods, lunch breaks, etc. They were allowed to choose their own rest periods and to give suggestions. Output increased in both the control rooms. It was concluded that social relationship among workers, participation in decision-making, etc. had a greater effect on productivity than working conditions

Relay Assembly Room 1

- *Manipulated factors of production to measure effect on output:*
  - Pay Incentives (Each Girls pay was based on the other 5 in the group)
  - Length of Work Day & Work Week (5pm, 4:30 pm, 4pm)
  - Use of Rest Periods (Two 5 minutes break)
  - Company Sponsored Meals (Morning Coffee & soup along with sandwich)

- *Results:*
  - Higher output and greater employee satisfaction

- *Conclusions:*
  - Positive effects even with negative influences – workers' output will increase as a response to attention
  - Strong social bonds were created within the test group. Workers are influenced by need for recognition, security and sense of belonging

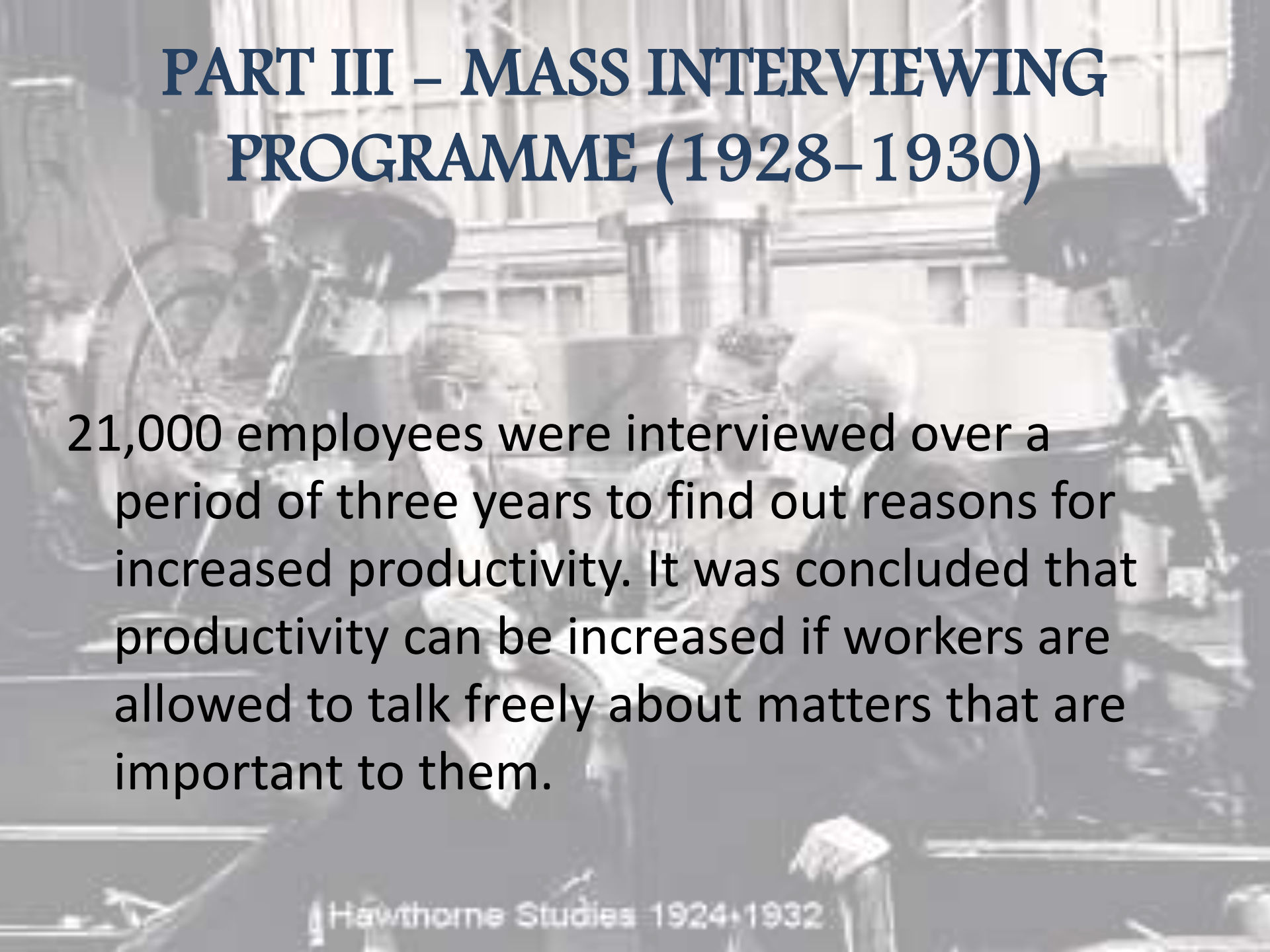
Relay Assembly Room 2

- *Measured effect on output with compensation rates*
  - Special observation room
  - 1st Session- Relay Assemblers changed from departmental incentive to small group incentive
  - 2nd Session - Adjusted back to large group incentive
- *Results:*
  - Small group incentives resulted in highest sustained level of production – 112% over standard output base
  - Output dropped to 96.2% of base in 2nd session
- *Conclusion:*
  - Pay relevant to output but not the only factor

WOMEN IN THE RELAY TEST ASSEMBLY ROOM

PART III – MASS INTERVIEWING PROGRAMME (1928–1930)

21,000 employees were interviewed over a period of three years to find out reasons for increased productivity. It was concluded that productivity can be increased if workers are allowed to talk freely about matters that are important to them.



- **Conducted 20,000 interviews.**

- **Objective was to explore information, which could be used to improve supervisory training.**

- **Initially used the method of Direct Questioning and changed to Non Directive.**

- ***Results:***

  - Merely giving an opportunity to talk and express grievances would increase the morale.

  - Complaints were symptoms of deep-rooted disturbances.

  -Workers are governed by experience obtained from both inside and outside the company.

- *Measured output with changes to work conditions only:*
  - Special Observation Room
  - Length of Work Day
  - Use of Rest Periods
  - Workers stayed on established Piece-rate compensation

- Result:
  - Productivity increased by 15% over standard output base

- *Conclusions:*
  - Productivity is affected by non-pay considerations
  - Social dynamics are a basis of worker performance

PART IV – BANK WIRING OBSERVATION ROOM EXPERIMENT (1932)

A group of 14 male workers in the bank wiring room were placed under observation for six months. A worker's pay depended on the performance of the group as a whole. The researchers thought that the efficient workers would put pressure on the less efficient workers to complete the work. However, it was found that the group established its own standards of output, and social pressure was used to achieve the standards of output.

- Limited changes to work conditions:
  - Segregated work area
  - No Management Visits
  - Supervision would remain the same
  - Observer would record data only – no interaction with workers

- Small group pay incentive

- Result:
  - No appreciable changes in output

- Conclusions:
  - Pre-existing performance norms
  - Group dictated production standards - Systemic Soldiering
  - Work Group protection from management changes.

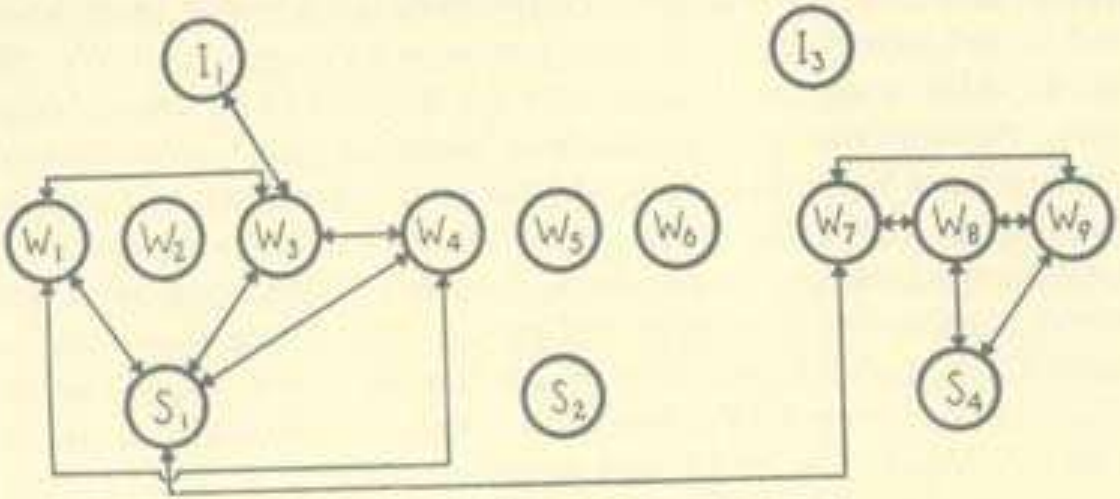


FIGURE 43
FRIENDSHIPS
BANK WIRING OBSERVATION ROOM

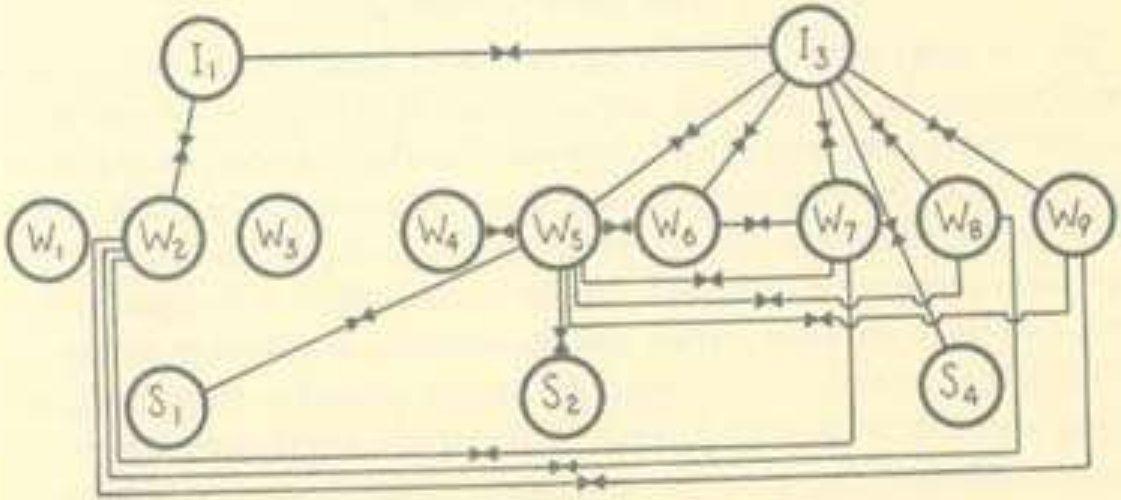


FIGURE 44
ANTAGONISMS
BANK WIRING OBSERVATION ROOM

CONCLUSIONS OF HAWTHORNE STUDIES / EXPERIMENTS

The conclusions derived from the Hawthorne Studies were as follows :-

- The social and psychological factors are responsible for workers' productivity and job satisfaction. Only good physical working conditions are not enough to increase productivity.
- The informal relations among workers influence the workers behavior and performance more than the formal relations in the organization.
- Employees will perform better if they are allowed to participate in decision-making affecting their interests.
- Employees will also work more efficiently, when they believe that the management is interested in their welfare.
- When employees are treated with respect and dignity, their performance will improve.
- Financial incentives alone cannot increase the performance. Social and Psychological needs must also be satisfied in order to increase productivity.
- Good communication between the superiors and subordinates can improve the relations and the productivity of the subordinates.
- Special attention and freedom to express their views will improve the performance of the workers.

CRITICISM OF HAWTHORNE STUDIES / EXPERIMENTS

The Hawthorne Experiments are mainly criticised on the following grounds :-

- **Lacks Validity** : The Hawthorne experiments were conducted under controlled situations. These findings will not work in real setting. The workers under observation knew about the experiments. Therefore, they may have improved their performance only for the experiments.
- **More Importance to Human Aspects** : The Hawthorne experiments gives too much importance to human aspects. Human aspects alone cannot improve production. Production also depends on technological and other factors.
- **More Emphasis on Group Decision-making** : The Hawthorne experiments placed too much emphasis on group decision-making. In real situation, individual decision-making cannot be totally neglected especially when quick decisions are required and there is no time to consult others.
- **Over Importance to Freedom of Workers** : The Hawthorne experiments gives a lot of importance to freedom of the workers. It does not give importance to the constructive role of the supervisors. In reality too much of freedom to the workers can lower down their performance or productivity.

THANK

YOU . . .